

A Robust Biometric Authentication System for Handheld Electronic Devices by Intelligently Combining 3D Finger Motions and Cerebral Responses

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Abstract—Rapid advancement in sensor technology through miniaturization of electronic components has enabled the consumer electronic (CE) research community including the manufacturers to embed various utility sensors into handheld devices. In addition to traditional sensors such as Inertial Measurement Unit (IMU), camera, fingerprint or proximity, futuristic sensors such as Electroencephalogram (EEG) or Electromyography (EMG) are also being included in the next-generation CE devices. Air or touch signature-based authentication systems are common in modern CE devices. However, cerebral activities clubbed with gestures will certainly enhance the security of such authentication systems. This can help consumers from being the victims of shoulder surfing attacks. In this article, a new method is proposed to verify air signatures by analyzing finger movements and cerebral activities together with the help of sensors in next-generation CE devices. Signatures are first spotted by analyzing 3D geometrical features of the finger movement during the signing. Concurrent EEG responses are then analyzed for the verification. Hidden Markov Model (HMM) and Random Forest (RF) classifiers have been used to train the system. Experiments reveal that EEG signals are highly correlated with the finger movements during air signatures even in the presence of motion artifacts. Therefore, false-positive rates have significantly reduced as compared to the existing tracking-based methods. Verification accuracy as high as 95.5% (HMM) and 98.5% (RF) have been recorded when tested on our dataset.

Index Terms—Biometric authentication, air signature, signature spotting, signature verification.

I. INTRODUCTION

ADVANCEMENT in sensor technology has enabled researchers to develop robust biometric systems capable of capturing air gestures for authentication and verification purposes [1] that can even be used in handheld devices. Moreover, the inclusion of miniaturized sensors in these devices has brought a revolution in the next-generation of consumer electronics (CE) products. Futuristic CE devices

will certainly include sensors such as EMG, ECG or EEG in addition to conventional sensors such as IMU, fingerprint or camera. These sensors help to measure physiological parameters during various activities. Therefore, the development of intelligent applications using the aforementioned advanced sensors can be started for the benefit of the users. Authentication and verification are very important but challenging features in CE handheld devices. Thus, the exploitation of advanced sensors for designing authentication protocol seems to be a timely activity.

Typical device authentication processes use biometric traits or features to verify whether a person is authorized to access the device or not. Air signature/gesture in 3D is a well-known biometric authentication process used in many setups. Air writing has the advantage of touch-less or paper-less authentication. Moreover, air signatures can be captured in 3D using motion sensors. Users can put signatures at random positions during air writing, enabling more security because it is difficult for the intruders to find the exact location of the signature. While signing in the air, users can randomly scribble before and/or after the actual signature. This makes it difficult to localize the signature. However, existing gesture-based authentication systems are prone to shoulder surfing attacks. Thus, incorporating additional information such as cerebral activities (through EEG responses) in the feature vector can make such systems robust to shoulder surfing attacks.

A. Motivation and Contributions

Air signatures are becoming very popular for biometric authentication. However, such systems are subject to identify theft if a professional forger observes or video-records the signing process. Therefore, we need to enhance the robustness of the signing process. One way can be to ask the signer to put the signature within a random writing. This will ensure that the forger does not get information about the start and end points of the actual signature. During the verification, first we need to localize or spot the signature in the random writing, followed by the actual verification. In other words, spotting is a problem that works at a coarser level and it does not defy the basic objective of biometric authentication. It can be stated that, despite erroneous spotting, verification can be achieved with higher accuracy if the feature-set is robust. Moreover,

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spotting needs to be fast since the processing is done across multiple temporal windows of the signal. Therefore, high-level features such as slope, direction, and curvature cannot be used as the process becomes time-consuming. This is the primary motivation to use simple but effective global feature that is less time consuming. However, the spotted regions may not fully contain the signature. Therefore, verification using the existing high-level features, as discussed earlier, may not be accurate. Thus, we used statistical information extracted from the EEG of the user for verification. To accomplish this, the following technical contributions are made:

- This article proposes a method to combine air signatures with cerebral activities (EEG responses) to implement a new biometric authentication process suitable for next-generation consumer electronic devices.
- A heuristic-based solution has been proposed to spot and localize the air signature segments in random writing.

The rest of the article is organized as follows. In Section II, we discuss the existing related works. The data acquisition and the feature extraction processes are described in Section III and Section IV, respectively. The training process adopted in this article is presented in Section V. Proposed verification methodology is described in Section VI. Results are presented in Section VII. Conclusion and future work are reported in Section VIII.

II. RELATED WORK

Kim and Hong [2] proposed a new multimodal biometric authentication approach using teeth image and voice as biometric traits to protect private information stored in the mobile devices. The method has an ERR of 2.13% over 1000 images. Liu and Song [3] proposed a real-time embedded finger-vein recognition system for authentication on mobile devices which takes about 0.8 seconds to verify one input finger-vein sample with an EER rate of 0.07% over a database of 100 subjects. Galka *et al.* [4] presented the concept of the embedded solution for voice biometric access system as well as automatic speech processing methods keeping performance optimization methods as one of the main concern. To overcome various challenges associated with iris acquisition on handheld imaging devices, Thavalengal *et al.* [5] suggested potential solutions for unconstrained acquisition using smartphones. Bao *et al.* [6] designed hand-gesture recognition network that can classify seven sorts of hand gestures in a user-independent manner and on real time. Lee [7] discussed security issue and advancement for human shoulder-surfing resistant personal identification number (PIN) entry methods used in ATMs, mobile devices, and electronic door locks. Papadopoulos *et al.* [8] proposed a PIN-based authentication method that operates on touchscreen devices. The idea behind this was to use hybrid images to blend two keypads with different digit orderings one for the user close to the device and another for the attacker who is looking at the device from a bigger distance. Sun *et al.* [9] proposed an authentication system PassMatrix, which is based on graphical passwords to resist shoulder surfing attacks. Basically, the authors tried to overcome the weakness of humans for choosing a bad password, to resist shoulder surfing attacks as well as multiple

camera-based attacks. Zhu *et al.* [10] tried to explore AI problems and captcha as a graphical password (CaRP) scheme to improving online security that are resistant to online guessing attacks, relay attacks and shoulder-surfing attacks and image hotspot problem. However, aforementioned methods suffer from a few problems, namely ease of use, shoulder surfing attacks, inadequacy in number of samples, etc.

To address some of the aforementioned problems, researchers proposed several improvements. For example, Shahzad *et al.* [11] proposed authentication scheme (known as BEAT) based on the user behavior of performing certain actions on the touch screens through gestures or signatures. Sengupta and Rathor [12] presented a design methodology of securing hardware accelerators against IP piracy and false claims of IP ownership. Labati *et al.* [13] proposed a touchless fingerprint recognition system based on the computation of 3D models. In [14], the authors further explored to the extension called Secure Drag-and-Type for securing the password entry against shoulder-surfing and spyware attacks. Vhaduri and Poellabauer [15] presented an implicit wearable device user authentication mechanism using combinations of three types of coarse-grain minute-level biometric features. These methods try to address shoulder surfing issues and easy of access.

More recently, Electroencephalography (EEG) and motion sensors are used independently as well as in combinations to design brain-computer interface (BCI) applications including biometric authentication and verification [16]–[21], gaming [22]–[24], physical rehabilitation [25], security [26], visual object classification through brain activities [27], marketing [28], 3D handwriting and signature recognition [29], gait-related brain activity detection [30], etc. However, other than the modalities, features and classifiers are also important. For examples, high-level features [29], 3D histogram of oriented trajectories (HOT) and histogram of oriented optical flow (HOOF) [31] are popular. To estimate the distance between the feature vectors, Dynamic time warping (DTW) [32] or minimal variance matching (MVM) [33] can be used. Classifiers such as Hidden Markov Model (HMM) [34], Support Vector Machine (SVM) [35], Neural Networks (NN) [36], [37], and Random Forest [38] are typically used for the classification.

Signatures can be imitated by imposters if they are allowed to observe the finger movements. To make signatures secure, people may be encouraged to sign in the air randomly. However, random writing makes a typical biometric setup complex. This is because, spotting of the signature is extremely important. In this article, a new approach has been proposed to spot signatures in random air writing and then verify by analyzing the EEG responses. As EEG response is unique, it can help to improve the robustness. Also, the signal can be captured conveniently using low-cost commercial sensors [39]. Therefore, such sensors are already in use with the handheld devices [40], [41].

III. DATA ACQUISITION

The dataset that has been used in this work for experiments, was prepared in two phases: (i) In training stage, we collected

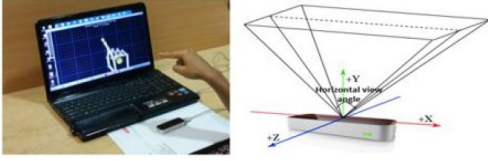


Fig. 1. A typical view of forward finger motion in the interaction space during air signature.

motion data as well as EEG recordings during only signature (without scribble). (ii) In the next stage, collection of motion sensor data and corresponding EEG responses were recorded. A motion sensor and an EEG sensor were used for acquiring the data. The motion sensor was used to collect index finger's movement during the signature. The EEG sensor was used to collect the cerebral activity of the user during air writing. The EEG device used in our work supports wireless connectivity with noise reduction techniques. The device is low cost as compared to other commercial EEG recording setups. Also, the headband EEG device does not require special liquid for recording the brain activities. This ensures that such devices can be used by mass and outside the laboratory environment. However, as the sensors used in our work are not used in everyday life, a convenient recording setup was prepared to guide the users. The users were given preliminary guidance to get acquainted with these sensors and their working principles.

A. Experimental Setup and Protocol

3D positions of the fingertips can be captured when they are placed within the field of view of the motion sensor. The sensor has a field of view similar to an inverted pyramidal structure with a range of 150° and it creates a $250 \text{ mm} \times 150 \text{ mm}$ interaction area as depicted in Fig. 1. A typical EEG setup with 14 electrodes as shown in Fig. 2, was used to record data and test its performance. Using both setups, physical movements and brain activities were recorded simultaneously. A set of guidelines were prepared and shared with the users to understand the experimental setup. First, the users were instructed about the working principles of these sensors. Next, the users were asked to put their hands within the field of view of the motion sensor. The users were asked to write in the air using their index finger while keeping other fingers closed as shown in Fig. 1. During the signature registration, EEG signals were also recorded using the setup shown in Fig. 2.

IV. FEATURE EXTRACTION

This section discusses the feature extraction process. There are two types of features (i) Global features of finger movement in 3D and (ii) statistical features extracted from the EEG data.

A. 3D Global Features

Finger motion in 3D is scaled within $[0,1]$ across three coordinate axes. A set of coarse-level features are extracted from the normalized data that represent the geometrical structure of the 3D trajectory representing finger movement. Geometrical

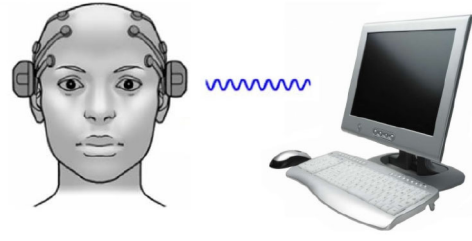


Fig. 2. A typical view of the EEG setup that has been used for recording cerebral activities.

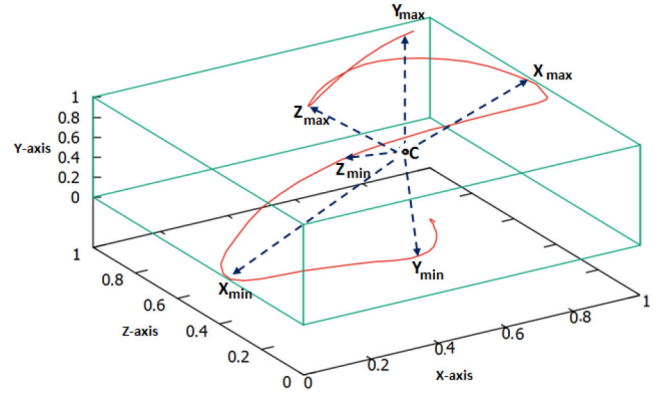


Fig. 3. A typical 3D orientation of motion data with various parameters used to construct the feature vector, where C represents the centroid of the data. a , b , and c represent the direction ratios of the line joining from the maxima or minima position to the centroid. d represents the normalized distance from the maxima or minima position to the centroid.

features f as defined in (1) are extracted from maxima and minima positions across all the axes as shown in Fig. 3,

$$f = [a, b, c, d] \quad (1)$$

In the aforementioned equation, a , b , c represent the direction ratios of the line joining from the maxima or minima position to the centroid of the 3D trajectory and d is the normalized distance from the maxima or minima position to the centroid of the track as shown in the Fig. 3. The trajectory can be represented using a 3D global feature vector F_{gl} that is the collection of feature-set f extracted from all six maxima or minima positions as denoted in (2).

$$F_{gl} = \begin{bmatrix} a_{x_{min}} & b_{x_{min}} & c_{x_{min}} & d_{x_{min}} \\ a_{x_{max}} & b_{x_{max}} & c_{x_{max}} & d_{x_{max}} \\ a_{y_{min}} & b_{y_{min}} & c_{y_{min}} & d_{y_{min}} \\ a_{y_{max}} & b_{y_{max}} & c_{y_{max}} & d_{y_{max}} \\ a_{z_{min}} & b_{z_{min}} & c_{z_{min}} & d_{z_{min}} \\ a_{z_{max}} & b_{z_{max}} & c_{z_{max}} & d_{z_{max}} \end{bmatrix}. \quad (2)$$

B. EEG Features

EEG signals may be corrupted by motion artefacts due to the body parts movement during air signature. Therefore, raw EEG signals have been smoothed using Savitzky Golay (S-Golay) filter as discussed hereafter.

Pre-Processing Using S-Golay Filter: Noisy EEG signals are first smoothed using a least square or polynomial-based approach called S-Golay Filter [42]. A low-degree polynomial

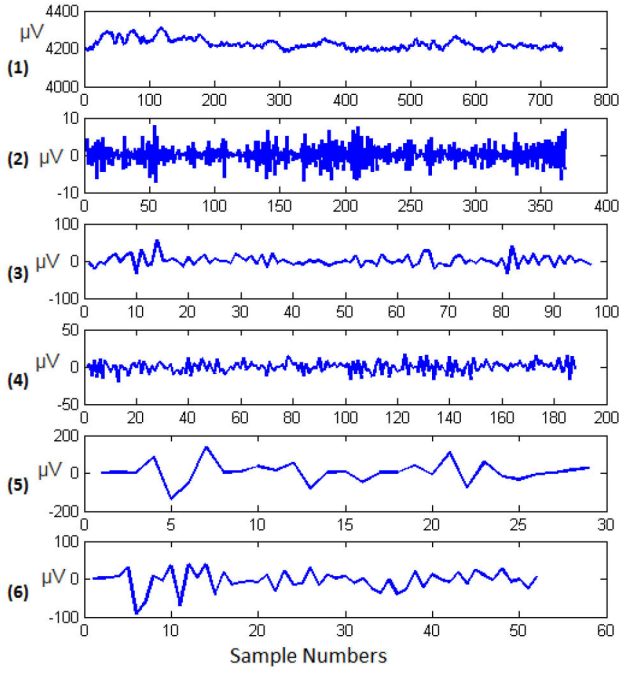


Fig. 4. 1. Raw signal 2. gamma band 3. beta band 4. alpha band 5. theta band and 6. delta band waves of decomposed DWT of one electrode.

is used to fit consecutive subsets of neighboring sample points. Let $S_i = f(t_i)$ be a signal with n samples. S-Golay filter can be applied using (3),

$$Q_i = \sum_{j=-\frac{p-1}{2}}^{\frac{p-1}{2}} c_j S_{i+j}, \quad \frac{p+1}{2} \leq i \leq n - \frac{p-1}{2} \quad (3)$$

where Q represents the smoothed signal, p denotes the frame span, and c_j represents the number of convolution coefficients. Values of c_j are computed using frame span p with the help of a quadratic polynomial. The main advantage of this approach is that, it ensures to preserve features of the distribution such as relative maxima, minima, and width that are usually 'flattened' by other averaging techniques such as moving average filter. We empirically set the windows size to 5 that worked fine on the dataset.

Extraction of Discrete Wavelet Transform (DWT) Features: DWT is a well-known approach for processing EEG signals. In DWT, multistage decomposition is used to convert an input signal into a series of small waves [43]. Decomposition of different frequency bands into approximation (A) and detail (D) coefficients is used to analyse a signal. Fig. 4 shows a schematic decomposition of an EEG signal, where Low pass filter (L) and High pass filter (H) have been used to filter the signals from noise. High-frequency fluctuations are removed from the signal and preserves slow trends using Low-pass filter (L). An approximation (A) of the signal are obtained by the outputs of low-pass filters. The slow trends are removed from the signal to preserve high-frequency fluctuations using High-pass filter (H). The wavelet coefficients that represent detail (D) information about the signal, are obtained using the output of the high-pass filter. The detail coefficients at next level can

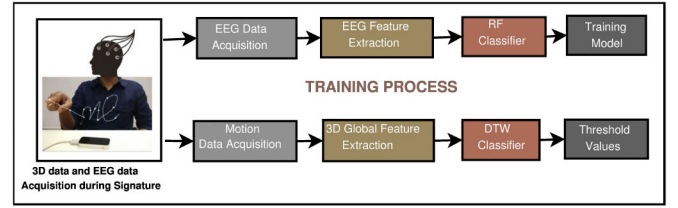


Fig. 5. A block diagram of typical training process.

be obtained by passing the approximation coefficients to L and H filters again and so on.

Four levels of EEG signal decomposition have been applied using the Daubechies 4 (DB4) wavelet decomposition technique. D_1, D_2, D_3, D_4 , and A_4 are the five coefficients that can correlate with the EEG spectrum of five different frequency bands popularly known as delta (1-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-22 Hz), and gamma (32-100 Hz), respectively. Gamma band waves are taken into feature construction as their variations have successfully been used for similar applications and analysis [44]. All 14 channels of EEG recordings are processed using the above analysis and standard deviation is estimated using (4),

$$SD = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (4)$$

where $x_i, \bar{x}$ and n represent the i^{th} signal, mean, and number of samples. Finally, a 14-dimensional feature vector is obtained as given in (5) corresponding to the gamma band wave,

$$F = \{e_1, e_2, e_3, e_4, \dots, e_{14}\} \quad (5)$$

where e_i represents the feature corresponding to the i^{th} electrode.

V. TRAINING PROCESS

During the training process as depicted in the Fig. 5, users register their signatures using fingertip movements by wearing the EEG device. 3D motion data and corresponding EEG data have been collected using the process discussed in Section III. Features have been extracted from the training data using the methods discussed in Section IV. HMM and RF classifiers have been used to train the classifiers. HMM has been used due to its ability to model temporal data. The classifier has been successfully used in speech, text and gesture recognition problems [29]. In our study, both EEG and 3D motion data are of temporal nature. Therefore, we used HMMs to model the sequential nature of the data. Similarly, Random Forest is a machine learning algorithm that works nicely on temporal data. It produces good results even without hyper-parameter tuning. Additionally, Random forests use the bagging technique which is based on averaging multiple deep decision trees. Parameters like window length W and sliding parameter L are estimated as described below.

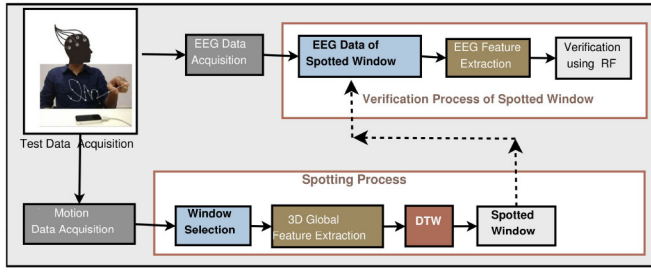


Fig. 6. A block diagram of the proposed signature spotting and verification processes.

Let $S^i = [s_1, s_2, s_3, \dots, s_m]$ be the set of m signatures registered by the i^{th} user. W and L are estimated using (6)-(7),

$$W^i = \frac{1}{m} \sum_{j=1}^m |s_j^i| \quad (6)$$

$$L^i = \frac{1}{m} \sum_{j=1}^m l_j^i \quad (7)$$

where $|s_j^i|$ is denoted as the number of sample points in s_j of the i^{th} user and l_j^i is the distance along x direction from the starting point to the local y -maxima point in the j^{th} signature of i^{th} user. Threshold values, namely D_{Max} and D_{Min} are estimated using DTW as described in (8)-(9),

$$D_{Max}^i = \text{Max}(D^1, D^2, \dots, D^m) \quad (8)$$

$$D_{Min}^i = \text{Min}(D^1, D^2, \dots, D^m) \quad (9)$$

where $D^i = \text{DTW}(s_p, s_q)$, and $\forall p, q \in m$ represent two indices of the signature samples such that $p \neq q$. D^i is the DTW distance between two samples s_p and s_q of the i^{th} user. These samples are represented using (2), e.g., $s_p = F_{gl}^p$ and $s_q = F_{gl}^q$.

VI. SPOTTING AND VERIFICATION OF SIGNATURES

The methodology is divided into two phases, namely signature spotting and signature verification as shown in the Fig. 6. In the first phase, probable locations of signatures (within random writing) are obtained using 3D global features extracted from the finger movement data acquired through motion sensor interface. A sliding window-based method has been considered to spot the possible locations of the signature within a random writing. In the next phase, the EEG data corresponding to the spotted locations are tested using HMM and RF for verification.

A. Spotting Using Global Features

Probable location of the signature within random air writing can be identified in the spotting phase. A sliding-window-based approach as shown in Fig. 7 has been adopted for spotting.

In the proposed sliding window-based approach, values of W and L estimated during the training phase, are used for re-positioning of the next window. To reduce computational overhead during the sliding process, the selection of the starting point of the next window is important. The selection

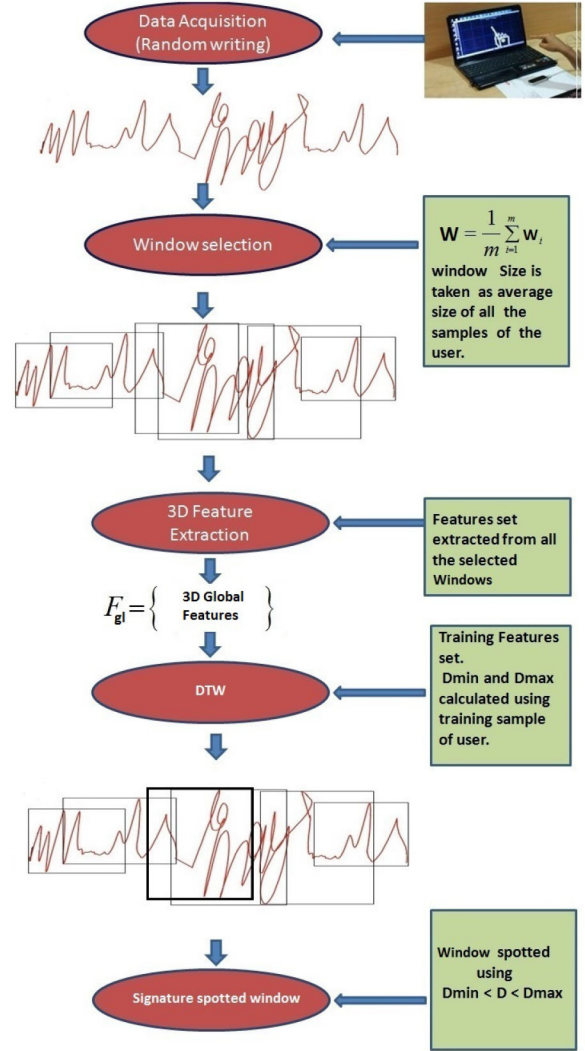


Fig. 7. Signature spotting using 3D global features and supervised classification.

process adopted in this work can be understood from the example given hereafter. Let x_1 and x_2 be the distances from the starting point of the window to the first local y -maxima and second local y -maxima points, respectively along the x -axis as shown in Fig. 8. Now, the distance from the starting point of the current window to the starting point of the next window (δx) can be estimated as follows:

$$\delta x = \begin{cases} x_2 - L, & \text{if } (x_2 - x_1) \geq L \\ x_1, & \text{otherwise.} \end{cases}$$

If there is no second local y -maxima toward the positive direction after the first local y -maxima within the current window, the next window starts from the position of the first y -maxima.

DTW has been used to find the matching score with the stored feature vectors in the training dataset. After processing every window, a few of the closely matching windows (within a predefined threshold) are considered as probable candidates that may contain the signatures. These windows are then processed for spotting the location of the signature. Assume n users are involved during the data collection process and each user has registered m signatures. Features of the training

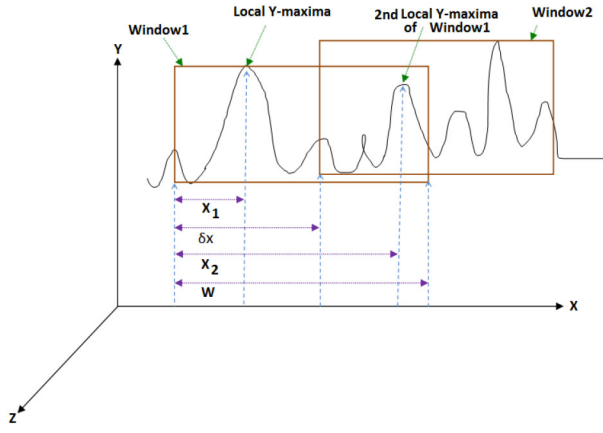


Fig. 8. A typical random writing with sliding window depicting various parameters.

samples are extracted using (2). Let $F_i = \{F_{gl}^1, F_{gl}^2, \dots, F_{gl}^m\}$ represents the set of feature vectors of m samples of the i^{th} user and s_{test} be a random writing sequence of a user who claims to be the i^{th} user. The aim is to spot the signature within s_{test} . Thus, every window of the test sample is compared with the training set of the i^{th} user using DTW matching score as given in (10),

$$D_j^i = DTW(F_{gl}^j, F_{test}^k) \quad (10)$$

where F_{gl}^j represents the feature vector of the j^{th} training sample of the i^{th} user and F_{test}^k is the feature vector of the k^{th} window of the given test sequence s_{test} . Average of D_j^i values for the k^{th} window is calculated using (11). This values is used against the thresholds (D_{min} and D_{max}) to validate the signature for verification.

$$AVGD_k^i = 1/m \sum_{j=1}^m D_j^i \quad (11)$$

Now, a decision to keep or reject the k^{th} window is taken based on the condition as mentioned in (12),

$$D_{min}^i < AVGD_k^i < D_{max}^i \quad (12)$$

where D_{min}^i and D_{max}^i represent the minimum and maximum distances obtained across all training samples of the i^{th} user as explained earlier. This is repeated for all windows of the test sequence and finally the window with the minimum distance is assumed to contain the signature of the user. If no candidate window is found satisfying the condition given in (12), it is assumed that the random writing does not contain any signature of the i^{th} user.

B. Verification of Signatures

Since the spotted window may not fully contain the signature, it often becomes difficult to verify the signature using the geometrical 3D features of that window. Therefore, the brain signals have been used for verification to reduce false rejection rates. EEG signals may be corrupted by motion artefacts due to the movement of the body parts during air signature. Therefore, raw EEG signals are smoothed using Savitzky

Golay (S-Golay) filter. After spotting using geometrical features, the EEG signals of the corresponding window are used in verification.

- *Verification Using HMM*: For EEG, gamma features are trained using HMM classifier. We created HMM models for all users using Baum-Welch training algorithm. $\theta = (T, O, \pi)$ represents each HMM model, where T denotes state transition matrix, O denotes observation matrix, and π denotes initial state probabilities [29], [34]. After training, a test window (spotted EEG window) is compared with all the models and measure probability of matching, i.e., likelihood. The model which gives the largest likelihood value, is assumed to be the user to whom the signature possibly belongs.
- *Verification Using RF*: The feature vector defined in (5) is then used for classification of a typical input EEG frame corresponding to the spotted signatures. A RF-based classifier has been used for classification. Random Forest is represented as the collection of classification trees, where each decision tree assigns the input data to one of the assigned classes [45].

VII. RESULTS AND DISCUSSIONS

This section describes the data as well as presents the results of spotting and verification. To the best of our knowledge, such multi-sensor datasets are not publicly available. Therefore, in our laboratory environment, we created a multi-sensor dataset involving 20 users with the help of motion and EEG sensors. We tested our approach using this dataset which we made available to the researchers.¹

A. Dataset Description

There is no publicly available multi-sensor dataset for the present task. Therefore, we made a setup in our lab environment to collect the air signatures as well as EEG responses. We involved 12 male and 8 female participants in the data collection. Half of the participants were below the age of 25 years and others were aged above 40 years. Each user was asked to provide 10 samples in two sessions (morning and evening). Motion sensor data and EEG signals were recorded simultaneously. The dataset has been used for training. In the next phase, these users were asked to provide signatures in air within some random writing as depicted in Fig. 9. Therefore, additional 200 samples were collected for testing. A summary of the dataset is presented in Table I.

B. Signature Spotting Results Using Global Features

Spotting results are presented first. A window is selected based on the analysis using global features when more than 60% of the segment within the window overlaps with the actual signature segment (ground truth). Such segments (windows) are considered as valid signatures as depicted in Fig. 10. User-wise spotting accuracy values are presented in Fig. 11. Overall accuracy of the spotting has been found to be 91.0%.

¹<https://sites.google.com/view/beherask/research>

TABLE I
SUMMARY OF THE DATASET USED

Type	Purpose	# of users	Samples/user	Total
3D signature	Training for Spotting	20	10	200
3D random writing	Testing in Spotting	20	10	200
EEG signal during signature	Training for verification	20	10	200
EEG signal during random writing	Testing in verification	20	10	200

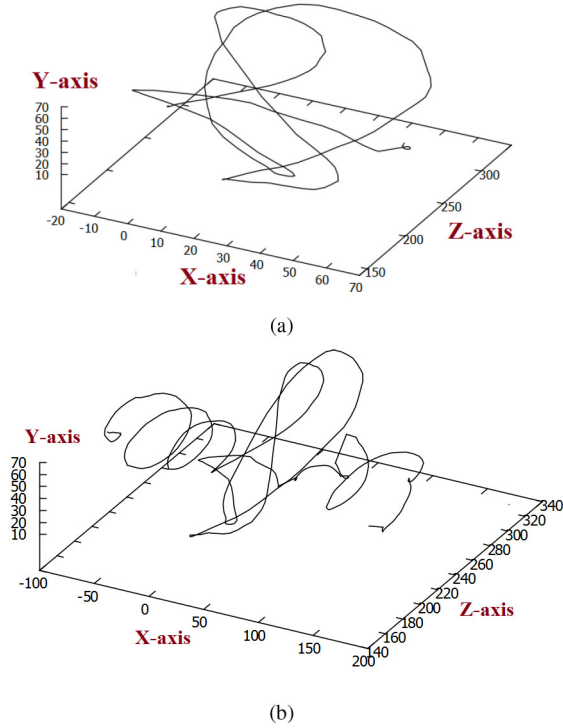


Fig. 9. (a) User's signature recorded using motion sensor. (b) User puts the signature within a random writing.

TABLE II
OVERALL VERIFICATION ACCURACY VALUES AND COMPUTATION OVERHEAD OF SPOTTED WINDOW USING VARIOUS FEATURES WITH THE HELP OF HMM AND RF

Dataset	Feature-Set	Accuracy (%)		Time (Sec.)	
		HMM	RF	HMM	RF
3D writing data	High level 3D feature	79.0	81.0	2.7	3.0
EEG data	EEG Statistical feature	95.5	98.5	2.0	2.5

C. Results of Verification Using EEG Features

Verification is applied on the spotted windows. EEG features corresponding to the time window of the spotted signature are fed to the classifiers. Fig. 12 presents a comparative results of verification using RF and HMM-based classifiers. Average verification accuracy using HMM has been found to be 95.5%. Average accuracy of verification has been found to be 98.5% when RF is used as the classifier. It reveals that RF outperforms HMM.

D. Discussion of Results

A summary of the results obtained using the proposed method is presented in Table II. Comparison of verification

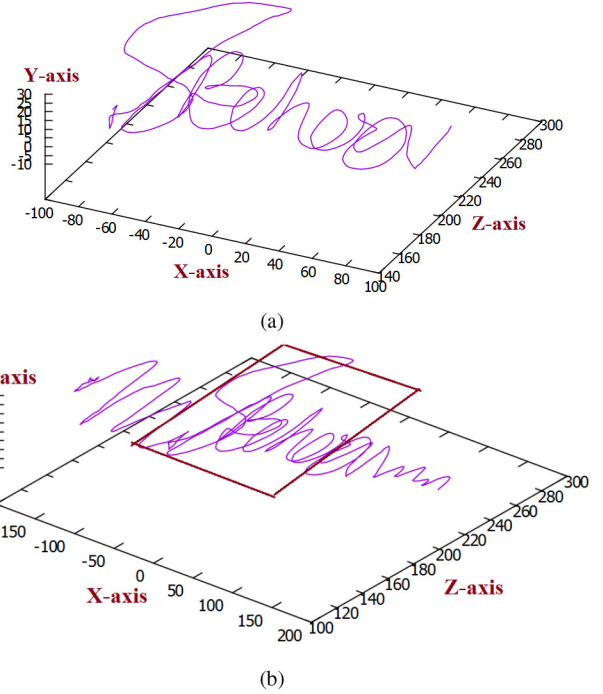


Fig. 10. (a) A typical window containing the signature of a user. (b) Spotted signature corresponding to the window shown in Fig. 10(a).

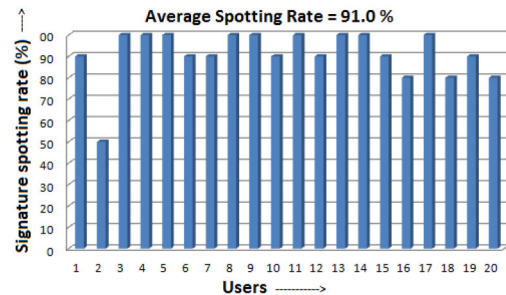


Fig. 11. User-wise signature spotting rate using 3D global features.

of a spotted window using EEG features with high-level features of 3D data has been presented in this table. It has been observed that verification using EEG features is better as the false rejection rate is reduced. It is unlikely that an imposter can produce random writing containing an imitated signature of a genuine user. Therefore, the false acceptance rate is very low. It has also been observed that verification using Random Forest is marginally better than HMM.

E. Analysis of Shoulder Surfing Attacks

In this section, we present a discussion on how the proposed method can be robust to shoulder surfing attacks. In our earlier

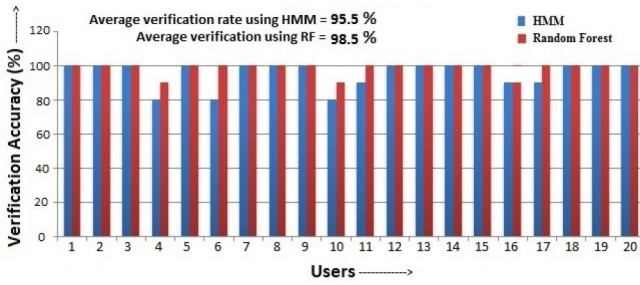


Fig. 12. Comparison of verification results obtained using HMM and Random forest.

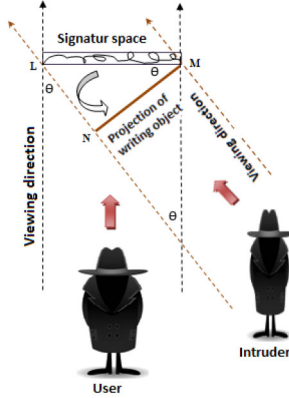


Fig. 13. Projections of a signature when viewed by two different persons from different angular positions.

work [1], we presented theoretically as well as through experimental evaluation that how shoulder surfing attacks can happen on gesture-based biometric systems. The present application is slightly different. Also, carrying out a similar experimental evaluation seems to be difficult at this stage due to the unavailability of trained attackers. Therefore, an in-depth theoretical analysis has been carried to demonstrate the robustness of the proposed two-stage biometric system against shoulder surfing attacks. We believe, the presented theoretical analysis can provide new insights into shoulder surfing attacks. The analysis is as follows:

Assume an intruder is trying to observe the signature by looking over the user's shoulder during the process of air signature. Due to the deviation in the projection line on the signature, the intruder gets a different view of the original signature. This essentially leads to a wrong pattern being watched by the intruder. The projection of the signature over the viewing plane creates a varying image of the signature as compared to the original signature. Therefore, different projections create different views on a signature. This can be explained using the image presented in Fig. 13. As depicted in the figure, there are two different persons (the user and the intruder) who are viewing the signature (contained within the straight line segment LM) from different angular positions (0° and θ°). As per the configuration, the intruder is viewing with an angle of θ° that creates the projection of the signature over the line segment NM . This can be estimated using (13).

$$NM = LM \cos \theta = \begin{cases} LM, & \text{if } \theta = 0^\circ \\ < LM, & \text{otherwise} \end{cases} \quad (13)$$

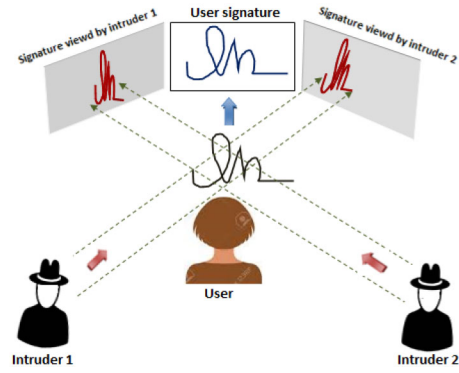


Fig. 14. Signature viewed by two intruders from different positions.

Since the original signature space ($\overline{LM}$) is projected onto a smaller line segment ($\overline{NM}$), some data points are expected to overlap. Hence the intruder gets an erroneous view of the signature. Moreover, intruders who view the signature from a wider angular position with respect to the user's projection line, are expected to accumulate more errors as the angle increases.

Similarly, when two intruders are trying to observe the signature from two different positions (with varying θ), they are expected to get erroneous views. Such a scenario is depicted in Fig. 14. If an intruder is able to find a position that is very close to the position of the user, the intruder may get a similar view of the user. However, due to the presence of randomness in the placement of the signature within the scribble, it is difficult for an intruder to accurately guess the start of the signature. Therefore, the Two-stage process proposed in this article is robust against shoulder surfing attacks.

F. Comparative Analysis

Since no work is available in the literature that uses fingertip motion and EEG responses together, we performed a comparison with the existing techniques that use motion sensors for air signature verification. Recently Guerra-Segura *et al.* [46] proposed an air-signature verification system using the same type of motion sensor used in our work. The authors used APIs of the sensor for extracting six types of features by dividing the time-series data into multiple segments. The verification was done using SVM with EERs of 0.25 and 1.25, respectively on random and expert forgers. However, in their study, all the forgeries were attempted by two imposters only and the verification was not attempted using any temporal classifiers. Likewise, the authors in [47] developed a sequential CNN architecture for geometric features extracted from air signatures and reported an average verification accuracy of 95%. We recorded accuracy as high as 95.5% and 98.5%, respectively using HMM and RF classifiers. We observed that our proposed multi-modal setup performs better in comparison to the aforementioned uni-modal systems. A comparison with other techniques is presented in Table III.

Providing an additional modality for user verification can certainly increase the system's robustness against various types of attacks. The proposed application has been designed keeping in mind the futuristic Augmented Reality (AR) or

TABLE III
COMPARATIVE ANALYSIS WITH EXISTING METHODOLOGIES

Authors	Methodology	Results
[46]	Mean, Correlation, Entropy, Kurtosis, etc. with SVM	0.25 EER 1.20 EER
[47]	Geometrical features and CNNs	95%
Proposed Work	Signature spotting, 3D geometrical features, DWT, HMM and RF	95.5% (HMM) 98.5% (RF)

Virtual Reality (VR) setups, where touch-less interactions are preferred. It is anticipated that such applications will be frequently used in next-generation devices as the users are rapidly adapting to such setups. Though attaching additional gadgets to the existing systems may create some discomfort to the users, particularly at the early stage of usage, however, our experience with the users involved during the training reveals that the level of discomfort decreases with the increasing number of trials. In futuristic designs, we are expecting that such interfaces will be commonplace. For example, the current extraordinary pandemic situation (COVID-19) has pushed humanity more towards touch-less interactions. Moreover, due to rapid advancement in such technologies, sophisticated devices are now being launched with the fewer interfaces (e.g., Muse² and Neurosky³ have lesser number of electrodes). Due to plug-and-play nature of connections with smartphones, these devices are good for large scale applications.

VIII. CONCLUSION AND FUTURE WORK

Modern sensor technology is enriching various research areas including human-computer interface and biometric. Moreover, the integration of these sensors with the next-generation handheld devices is being adopted at a rapid speed. This article proposes a method to spot air signatures scribbled within the random writings recorded through portable motion sensors embedded in CE devices and verify these signatures by analyzing corresponding EEG responses. EEG responses are recorded using easy-to-use low-cost sensors attached with such devices. The classification has been performed using HMM and RF classifiers. In the first phase, the signatures are spotted using a heuristic that relies on the 3D geometrical features extracted from the raw data captured through the motion sensor attached with the device. In the next phase, spotted signatures are verified using HMM and RF classifiers. Both classifiers perform satisfactorily with RF has a slight edge over the HMM. The method can be highly effective in enhancing the capabilities of typical CE setups. In the future, advanced sequential learning methods such as temporal CNNs or LSTMs can be used for improving the verification results.

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²<https://choosemuse.com/compare/>

³<http://neurosky.com/>

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